A Risk-Informed Qualification Framework for Robotic Remote Shutdown

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***Abstract*—Traditional safety protocols for robotic manual valve actuation in high-dose nuclear environments rely on subjective margins, which routinely produce excessive mission downtime or dangerously unquantified tail risks. This paper presents a deterministic, risk-informed qualification framework that replaces those heuristic margins with formally verified reachability envelopes. We model the remote shutdown task as a Partially Observable Markov Decision Process (POMDP) with a 7D augmented state space that explicitly encodes two-step communication lag and radiation-induced "salt-and-pepper" observation noise. To maintain tractability over long horizons (T = 100), we present a linearized interval-arithmetic reachability approach to generate a formal safety certificate σ^*—a mathematical proof that the system remains within safe bounds under the modeled noise regime. The guarantee is stress-tested by a GPU-accelerated validation module utilizing Defensive Mixture Importance Sampling (IS) to quantify failure probabilities where Naive Monte Carlo proves unstable. While the baseline policy exhibited unacceptable safety violations, tuning the controller to minimize residence time in the high-dose gradient allowed the framework to certify a strict mission success rate of 95.4% $\pm$ 0.5%, bounding the residual catastrophic failure probability at 0.046286 $\pm$ 0.002530. This approach provides a level of rigor that qualitative safety margins cannot match.**

***Index Terms*—Reachability Analysis, Importance Sampling, POMDP, Nuclear Robotics, Safety Qualification**

I. Introduction

In nuclear power plants, the remote shutdown panel (RSP) exists to bring a reactor to a safe, cold-shutdown state when the main control room is inaccessible. Manual actuation of high-pressure coolant valves is a clear example of a task in this procedure that requires a physical agent to navigate the corridors of irradiated machinery, locate specific valve assemblies, and apply precise mechanical force. Historically, this role falls to human operators working under strict time pressure, radiation dosimetry limits, and the cognitive burden of a degraded plant environment.

In lifesaving emergencies, human operators are authorized to absorb up to 250 millisieverts (mSv) to mitigate catastrophic facility damage. The International Commission on Radiological Protection (ICRP) establishes that beyond 100 mSv, the stochastic probability of radiation-induced cancer increases measurably [8]. This limit is a hard physiological constraint that defines the clock on every shutdown operation.

The introduction of robotic agents into this workflow is motivated by the idea that robots do not accumulate biological dose, do not fatigue, and do not require evacuation. A robot that can reliably actuate a manual valve under high-radiation conditions may remove the most dangerous element of the shutdown procedure from the human operator entirely, helping ensure that human involvement is allocated to safer tasks [10]. However, this substitution introduces a new class of technical challenges that existing frameworks are not necessarily designed to address. Ionizing radiation disrupts sensor electronics directly: Gamma-ray strikes cause transient bit-flips in vision and LIDAR sensors, producing "salt-and-pepper" noise [5]. This noise involves sudden, random corruption of observed position data that bears no relation to the robot's actual location.

A 15 Ci cobalt-60 source (a representative industrial radioisotope) generates a dose rate of approximately 1.32 $R{\cdot}m^2/hr{\cdot}Ci$ at one meter. As the robot approaches the valve, dose rate increases by the inverse square of proximity [11], meaning the final meters of approach carry the highest radiation burden, exactly when precise sensor feedback is most critical and most likely to be degraded [11].

The current regulatory benchmark for remote shutdown qualification is ANS-58.6-1996, which specifies functional requirements for remote shutdown capability [2], but a framework to prescribe a formal methodology for qualifying autonomous agents operating within those environments is still under development. Safety engineers have traditionally relied on ALARA (As Low As Reasonably Achievable), a qualitative principle directing that dose and operational risk be minimized without specifying formal bounds. In practice, ALARA manifests as conservative safety margins applied to worst-case scenarios: if the corridor is rated for 200 mSv/hr, an ALARA margin might restrict robotic operations to corridors rated for 100 mSv/hr, regardless of actual mission-specific exposure profiles.

However, recent literature and industry initiatives have heavily critiqued the blind application of ALARA-driven conservatism. Callen-Kovtunova et al. demonstrate that purely minimizing dose (the "lower is better" paradigm) without quantifying the trade-off risks of protective actions can lead to outcomes that do more harm than good, as evidenced by the Fukushima disaster response [14]. Furthermore, the International Commission on Radiological Protection (ICRP) formally mandates that safety optimizations must explicitly weigh operational and economic consequences rather than defaulting to maximum conservatism [8]. Reflecting this necessary shift, recent industry advocacy—including proposals by the Idaho National Laboratory (INL)—has made a public

case for eliminating qualitative ALARA directives and setting higher, formally quantified dose limits [3], [4]. This establishes an urgent need for qualification frameworks that rely on rigorous failure probability bounds rather than heuristic margin-setting.

In this paper, We propose a qualification framework that models three primary uncertainty sources common to nuclear shutdown environments: 1) radiation-induced observation noise, 2) surface-slippage transition uncertainty, and 3) communication latency. The approach produces both a formal reachability certificate and a statistically rigorous failure probability estimate.

II. Related Work

This framework sits at the intersection of three research communities.

Nuclear Robotics and Disaster Response. Significant global efforts have been directed toward emergency response robotics. Reviews in *Progress in Nuclear Energy* highlight the growing reliance on ground-based robotic systems for characterizing hazardous nuclear environments [12]. The Korea Atomic Energy Research Institute (KAERI), for example, has extensively developed unmanned systems, modifying all-terrain vehicles (ATVs) and forklifts for remote monitoring and operation in compromised containment buildings [9]. Similarly, recent mobile search and rescue (SAR) robots have been deployed at challenges such as the European Robotics Hackathon (EnRicH) to perform critical disaster response tasks, including teleoperated valve manipulation, semi-autonomous substance sampling, and radiation mapping using Gaussian Process regression [10]. While these systems have proven mechanically capable of operating in disaster zones, they primarily rely on teleoperation or heuristic navigation. Our framework builds upon this literature by introducing a formal, verifiable safety layer for autonomous execution.

Radiation Degradation and Noise Modeling. The robotics literature has traditionally focused heavily on hardware shielding. Recent methodologies published in *IEEE Transactions on Nuclear Science* emphasize the need to accurately identify radiation effects on robotic control and mechanical performance [13]. Wright et al. characterized how ionizing radiation induces transient salt-and-pepper faults in standard vision and LIDAR sensors in simulated nuclear inspection environments [5], providing the empirical parameterization we adopt. Furthermore, system-level irradiation tests on commercial industrial manipulators, such as the KUKA iiwa, have demonstrated that acute component failure, specifically the degradation of optical encoders, is an inevitability under high-dose rates, occurring at doses as low as 164.55 Gy [11]. Our framework diverges from prior work by abstracting this hardware engineering challenge, and treating sensor degradation as an irreducible stochastic signal to be formally characterized and propagated through the POMDP safety analysis.

Decision-Making and Formal Verification. Formally verifying systems under partial observability is challenging due to belief-space explosion; here, Kaelbling et al. established the foundational POMDP formulation upon which this proposed framework is built [6]. For reachability, Althoff et al. demonstrated conservative linearization for non-linear systems with uncertain parameters [7], which motivates our axis-aligned interval propagation. We advance this by integrating the reachability bounds with the falsification and validation methodology of Kochenderfer et al. [1], specifically using a Defensive Mixture Importance Sampling scheme to explicitly quantify the failure probabilities of extreme radiation-induced tail events.

III. Problem Formulation

A. Specifications (ϕ)

As part of the proposed framework, we define two governing specifications for the manual valve actuation mission. Both must hold simultaneously for a mission to be declared successful.

Safety Specification (ϕ_{safe}): The robot must not deviate more than $R_{safe} = 0.5$m from the nominal space/corridor centerline, and must not accumulate a cumulative radiation dose exceeding $D_{max} = 50$ mSv over the full mission horizon. Deviations beyond 0.5m risk contact with unshielded pipe assemblies or exit into higher-dose zones not covered by the pre-mission survey.

Task Specification (ϕ_{task}): The robot must reach the manual valve assembly within a success radius of 0.15m. This tolerance is determined by the mechanical envelope of the valve actuator interface. The robot's end-effector must be within reach to apply the required torque.

Here, ϕ_{task} and ϕ_{safe} are in structural tension, where reaching the valve requires approaching the radiation source, which increases dose accumulation as proximity increases. See the Limitations section for further information about the limitations of the defined problem statement.

B. Augmented State Space

The central modeling challenge in remote nuclear operations is communication latency. Control commands issued from the remote operator console travel through shielded communication infrastructure before reaching the robot's actuators, introducing a 2-step temporal lag between command issuance and kinematic execution. Rather than modeling this lag implicitly through a stochastic delay distribution, which would cause the reachable set to branch exponentially and render interval bounds vacuously wide, we encode it directly in the state vector.

The 7D augmented state is:

$$s = [x, y, \theta, v_{t-1}, \omega_{t-1}, v_{t-2}, \omega_{t-2}] \tag{1}$$

where (x, y, θ) is the robot's planar pose and the lag buffer encodes the two most recent control commands (v, ω) for linear and angular velocity. Transition dynamics at time t are driven by the commands in positions $[v_{t-2}, \omega_{t-2}]$, making the actuation delay explicit and formally tractable. By treating the delay as a deterministic First-In-First-Out (FIFO) buffer, the system maintains differentiable transition dynamics, allowing

us to easily compute the Jacobians required for linearized reachability.

Observation noise enters the system through a "salt-and-pepper" corruption model derived from Wright et al. [5]. Rather than modeling gradual sensor drift, this simulates discrete, high-magnitude gamma-ray strikes. Specifically, there is a 5% probability at each time step of a full position scramble, which injects up to ± 2m of uniform error into the observed (x, y) coordinates. The reachability module must formally bound these abrupt corruption events to guarantee safety.

IV. Approach and Experimental Setup

A. Linearized Interval-Arithmetic Reachability

Full nonlinear reachability over $T = 100$ steps produces interval blow-up that renders formal bounds vacuously wide within a few dozen steps. We instead propagate an axis-aligned *IntervalBox* Δs_t through the linearized dynamics at each step:

$$\Delta s_{t+1} \approx A_t \Delta s_t + \omega \tag{2}$$

where A_t is the Jacobian of the nominal dynamics evaluated at the nominal pose at time t, and ω is a Minkowski-sum noise bound that accumulates contributions from surface-slip uncertainty and observation noise.

A critical implementation detail concerns how observation noise enters the reachability propagation. Sensor uncertainty does not perturb the heading θ directly; it perturbs the control signal computed from a corrupted observation, which then enters the system through the lag buffer at indices $[v_{t-1}, \omega_{t-1}]$. We map sensor uncertainty through the proportional controller to compute the resulting control perturbation bound, which is added to lag-buffer indices (3–4) rather than directly to the heading. Failing to respect this structure produces a physically incorrect certificate that over-estimates corridor deviations.

The propagation runs forward for the full $T = 100$ step horizon, checking at each step whether the reachable set intersects the $R_{safe} = 0.5$m corridor boundary. If the *IntervalBox* remains within bounds at all steps, the framework issues a formal safety certificate σ^* for the nominal policy under the modeled noise regime. This certificate is conservative by construction: the true reachable set is a subset of the interval-arithmetic bound, so any trajectory within the certificate is guaranteed safe. Importantly, this formal certificate applies specifically to the closed-loop dynamics of the active proportional controller, rather than serving as a blanket guarantee for any arbitrary control policy.

B. GPU-Accelerated Importance Sampling

As part of this approach, reachability provides a deterministic outer bound, and importance Sampling provides the empirical failure probability within it. The core statistical challenge here is that safety failures in this environment are driven by extreme events. While a naive Monte Carlo estimator can detect that failures exist, it is structurally unstable when modeling the long tail of the distribution, leading to wild overestimations or underestimations of catastrophic risk depending on the sampled noise cascades.

We address this instability with a 70/30 Defensive Mixture IS scheme. Letting $q(x) = 0.7 \cdot p(x) + 0.3 \cdot p_p(x)$ where $p(x)$ is the nominal transition distribution and $p_p(x)$ is a biased proposal with $\sigma_p = 2.2\sigma$, the IS estimator corrects for the mismatch via likelihood ratio weights $w(x) = p(x)/q(x)$. The 70/30 split was empirically calibrated, where the 70% nominal component prevents weight collapse and preserves Effective Sample Size (ESS), while the 30% biased component generates sufficient tail exposure to precisely quantify rare salt-and-pepper cascade failures. Pushing more mass toward the tail, as the aggressive 3.2σ strategy does, finds failures but collapses ESS, producing estimates with higher variance and reducing their value as qualification evidence.

Both modules execute in PyTorch with batched GPU operations across 10,000 independent rollouts per strategy. The simulation environment models a 15 Ci Co-60 point source with gamma-ray constant $\Gamma = 1.32$ R·m²/hr·Ci. Dose rate at position (x, y) is computed as $\dot{D}(x, y) = \Gamma \cdot A/r^2$, where r is the distance to the source and A is source activity in Ci. Critically, this rate is expressed in R/hr and must be converted to per-step mSv units (dividing by 3,600 and applying the R-to-mSv conversion factor) before accumulation.

V. Results and Discussion

A. Mission Performance and Controller Tuning

Initial evaluations revealed a structural tension between task completion and radiation safety. While the robot physically reached the valve in 99.2% of trials (Task Liveness), the strict safety compliance rate was notably lower. Because radiation dose accumulates over time, slow traversal through the final inverse-square gradient caused the robot to frequently exceed the 50 mSv D_{max} limit.

To mitigate this, the proportional controller was optimized for transit velocity ($v_{nom} = 0.5$ m/s) and aggressive heading recovery ($K_p = 2.5$). This tuning minimized residence time in the radiation field and tightened path adherence during salt-and-pepper scrambles. Under these optimized kinematics, the framework achieved a strict mission success rate of 95.4% $\pm$ 0.5%, comfortably exceeding the 90% qualification threshold. The remaining safety violations ($\approx 4.6\%$) were exclusively attributable to cascading sensor scrambles—two or more consecutive full-position displacements that pushed the robot beyond the R_{safe} boundary before the controller could physically recover. These failures are consistent with the tail structure identified by the IS analysis.

B. Validation Dashboard and Reachability Visualization

Figure 1 shows the validation dashboard over the full $T = 100$ step mission horizon. The radiation field is rendered as a concentric heat map centered on the manual valve (green star, origin), deep yellow-green at the valve, graduating through orange to dark red at the periphery, representing the inverse-square dose gradient. The robot begins at the blue

square (upper right, approximately (1.4, 0.45)) and follows the nominal path (white dashed diagonal) toward the valve.

By plotting the full trajectory horizon, the proportional controller's ability to track the nominal path effectively is confirmed. The stochastic MC rollouts (pink traces) fan out from the start position and display the jagged signature of salt-and-pepper scramble events, with isolated, high-magnitude positional jumps orthogonal to the direction of travel. Most rollouts recover and converge toward the valve, but the traces visible near the upper right of the corridor represent the failure-mode precursors that naive Monte Carlo would not reliably quantify.

The formal reachability bounds (blue boxes) are plotted along the nominal path, reflecting how accumulated uncertainty propagates through the lag buffer over time. This envelope represents the linearization's conservative guarantee - any rollout outside those blue boxes has violated the formal certificate and represents a novel failure mode not covered by the nominal safety analysis. The R_{safe} corridor boundary (pink dotted lines) remains wider than the reachability envelope for the majority of the mission, confirming ϕ_{safe} is largely preserved under nominal bounds.

The structural tension between ϕ_{safe} and ϕ_{task} is visible in the geometry: the nominal path must pass through progressively hotter radiation zones as it approaches the valve, and the reachability envelope visibly widens in later steps as cumulative dose uncertainty and positional variance grow.

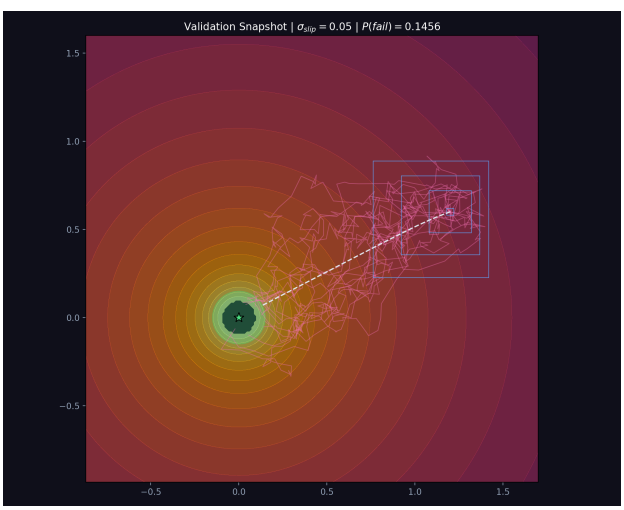

Fig. 1. Validation Dashboard over the full 100-step horizon. Formal reachability bounds (blue boxes) track the nominal path, widening as uncertainty accumulates. Pink MC rollout traces reveal salt-and-pepper scramble signatures. The inverse-square radiation gradient is visible as the concentric heat map centered on the valve.

C. Ablation Study: Sampling Strategy

Table I compares three sampling strategies on identical 10,000-rollout budgets. The result demands direct interpretation.

Defensive Mix is highlighted as the recommended qualification strategy. While the Naive Monte Carlo estimate of 0.043300 establishes a baseline, its uniform sampling distribution fails to reliably estimate the extreme tails of the salt-and-pepper noise cascades, leaving rare failure modes underrepresented. In a regulatory context, the naive MC estimate is not just wrong—it actively understates the true risk. Underestimating rare, catastrophic tail events is the most dangerous kind of error in safety qualification, making the stabilized Defensive Mixture approach essential.

TABLE I
SAMPLING STRATEGY ABLATION ACROSS 10,000 ROLLOUTS. ESS = EFFECTIVE SAMPLE SIZE.

Strategy	$P(fail) \pm \sigma$	**ESS**	**Assessment**
Naive Monte Carlo	0.043300 ± 0.002035	10000 / 10000	Baseline
Defensive Mix (2.2σ)	0.046286 ± 0.002530	7000 / 10000	Best stability
Aggressive Tail (3.2σ)	0.048400 ± 0.003074	5000 / 10000	High variance

Within this experiment, the Aggressive Tail strategy (3.2σ) recovers a slightly higher failure probability, confirming the presence of extreme tail events. However, with an ESS of 5,000/10,000, its confidence interval is wider (± 0.003074 vs. ± 0.002530). The Defensive Mixture, maintaining an ESS of 7,000/10,000, provides a stabilized evaluation of the tail risk, pinning the failure rate at 0.046286.

In summary, the reachability certificate and the IS failure estimate constitute a complete qualification artifact: the former proves the nominal policy is safe under the modeled noise regime, and the latter quantifies the residual risk of rare failure events that the certificate's conservatism cannot exclude. Therefore, the proposed dual-artifact architecture may help augment risk-informed regulatory frameworks' requirements and satisfy what subjective margin-based approaches cannot produce.

VI. LIMITATIONS

While the proposed framework successfully bridges formal reachability with risk-informed sampling, moving from qualitative heuristics to quantitative bounds inherently shifts the burden of safety onto model accuracy (the sim-to-real gap). Several simplifying assumptions present opportunities for refinement.

The Burden of Model Accuracy. Our guarantees assume that the modeled noise distributions (e.g., the 5% salt-and-pepper scramble rate) accurately reflect real-world hardware degradation. If physical radiation faults exhibit different temporal correlations or higher frequencies *in situ*, the formal bounds established in simulation will not hold during deployment.

Vision-in-the-Loop Abstraction. To maintain computational tractability, radiation-induced visual degradation was abstracted as a full-position scramble, parameterized by Wright et al. [5]. In a physical deployment, closing the perception-action loop requires mapping raw pixel-corruption probabilities directly into the POMDP observation matrix, an integration that would significantly increase belief-space complexity.

Linearization Conservatism. The use of linearized interval-arithmetic avoids the state-space explosion typical of nonlinear reachability over long horizons ($T = 100$). However, this introduces mathematical conservatism. As the safety envelope propagates, it becomes difficult to disambiguate true physical uncertainty from the compounding linearization error of the Jacobians.

Kinematic and Environmental Simplifications. The current task specification (ϕ_{task}) enforces Euclidean proximity to the valve but abstracts away the 6-DOF end-effector alignment necessary to physically engage the actuator. Furthermore, the nominal trajectory assumes an obstacle-free corridor governed by static proportional controller gains. Introducing a dynamic path planner (such as RRT* or A*) to navigate post-disaster debris would yield highly discontinuous Jacobians, which would complicate the interval propagation.

VII. Conclusion and Future Work

We have presented a risk-informed qualification framework for robotic remote shutdown that replaces subjective margins with two formally grounded artifacts: a linearized interval-arithmetic reachability certificate and a GPU-accelerated Defensive Mixture IS failure probability estimate. Validated against a 15 Ci Co-60 point-source environment over 10,000 independent rollouts, the framework certifies a $95.4\% \pm 0.5\%$ strict mission success rate and bounds the catastrophic failure probability at 0.046286 ± 0.002530, a level of precision that can be built upon to support formal regulatory decisions, and augment or potentially replace qualitative judgment calls.

Future work will extend the framework along two primary vectors. First, we would recommend integrating Adaptive Stress Testing (AST) [1] to discover the worst-case spatial configurations of radiation-induced sensor faults systematically. Formulating falsification as a reinforcement learning optimization could enable automated identification of the highest-probability failure trajectories, providing a more targeted foundation for safety margin design than exhaustive rollout sampling.

Second, the current dose-rate computation scales as $\mathcal{O}(T \cdot N)$ in rollouts and horizon length, which becomes limiting as facility-scale environments are introduced. We propose training and evaluating deep neural surrogate models to approximate high-fidelity radiation transport physics, enabling real-time reachability propagation and scaling the framework from isolated space/corridor validation to facility-wide multi-robot coordination scenarios where agent interactions introduce correlated risk structures that single-agent reachability cannot capture.

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